

# Probability - Set 1 - Solutions

Specification M7.1-M7.7.

## Answer key

| Question | Answer | Topic       |
|----------|--------|-------------|
| 1        | D      | Probability |
| 2        | C      | Probability |
| 3        | F      | Probability |
| 4        | A      | Probability |
| 5        | C      | Probability |
| 6        | F      | Probability |
| 7        | D      | Probability |
| 8        | A      | Probability |
| 9        | E      | Probability |
| 10       | A      | Probability |
| 11       | D      | Probability |
| 12       | A      | Probability |

## Worked solutions

### Question 1 PRO-01 · Probability · spec M7.7

A bag contains 4 red and 6 blue counters. Two counters are taken at random without replacement. Find the probability that they are different colours.

**Answer: D**

$$P(RB) + P(BR) = \frac{4}{10} \cdot \frac{6}{9} + \frac{6}{10} \cdot \frac{4}{9} = \frac{48}{90} = \frac{8}{15}.$$

#### Full method - how and why

'Different colours' happens in two orders. Either add the two ordered cases or use the complement.

1 Red then blue:  $\frac{4}{10} \times \frac{6}{9} = \frac{24}{90}$ .

why: After removing a red, 9 counters remain and all 6 blues are still there - **without replacement**.

2 Blue then red:  $\frac{6}{10} \times \frac{4}{9} = \frac{24}{90}$ .

why: The same product, but the orders are distinct outcomes so both count.

3 Add:  $\frac{48}{90} = \frac{8}{15}$ .

why: Mutually exclusive cases add; then cancel by 6.

4 Check by complement:  $P(\text{same}) = \frac{4 \cdot 3}{10 \cdot 9} + \frac{6 \cdot 5}{10 \cdot 9} = \frac{12+30}{90} = \frac{42}{90}$ , and  $1 - \frac{42}{90} = \frac{48}{90} \checkmark$ .

why: Two routes agreeing is strong confirmation.

#### Where it goes wrong

- Counting only one order, giving  $\frac{4}{15}$  - an offered option.
- Keeping the denominator at 10 for the second draw ( $\frac{12}{25}$ ), which is the **with replacement** answer.

**Takeaway** Without replacement: reduce both the relevant colour count and the total; and count both orders unless the question fixes one.

### Question 2 PRO-02 · Probability · spec M7.6

Two fair dice are thrown. Find the probability that the product of the two scores is odd.

**Answer: C**

The product is odd only if both scores are odd:  $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$ .

#### Full method - how and why

A product is odd only when **every** factor is odd - so this is an 'and' of two independent events.

1 Odd  $\times$  odd = odd; any even factor makes the product even.

why: Recognising this converts the question into a much simpler one.

2  $P(\text{first die odd}) = \frac{3}{6} = \frac{1}{2}$ ; same for the second.

why: Three odd faces out of six: 1, 3, 5.

3 Independent dice, so multiply:  $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$ .

why: One die's outcome has no effect on the other.

#### Where it goes wrong

- Answering  $\frac{1}{2}$  by thinking 'half of all products are odd' - in fact only 9 of the 36 outcomes are.
- Treating it as a **sum** (where odd + odd = even), a different and much more common question.

**Takeaway** Product odd  $\Leftrightarrow$  all factors odd; product even  $\Leftrightarrow$  at least one even.

### Question 3 PRO-03 · Probability · spec M7.7

$P(A) = 0.6$ ,  $P(B) = 0.5$  and  $P(A \text{ and } B) = 0.3$ . Which of the following statements is/are true?

- I  $A$  and  $B$  are independent.
- II  $A$  and  $B$  are mutually exclusive.
- III  $P(A \text{ or } B) = 0.8$

**Answer: F**

$P(A)P(B) = 0.3 = P(A \text{ and } B)$ , so independent (I). Not mutually exclusive since they can both happen (II false).  
 $P(A \text{ or } B) = 0.6 + 0.5 - 0.3 = 0.8$  (III).

#### Full method - how and why

Three definitions to test: independence, mutual exclusivity and the addition rule.

- 1 I: independence means  $P(A \cap B) = P(A)P(B)$ . Here  $0.6 \times 0.5 = 0.3$ , which matches. **True.**  
why: Independence is a numerical condition, not something you can eyeball.
- 2 II: mutually exclusive means  $P(A \cap B) = 0$ . It is  $0.3 \neq 0$ . **False.**  
why: They overlap, so they can happen together.
- 3 III:  $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.6 + 0.5 - 0.3 = 0.8$ . **True.**  
why: Subtract the overlap or you count it twice.
- 4 So I and III only.  
why: Match to the list.

#### Where it goes wrong

- Treating independent and mutually exclusive as the same thing - they are near-opposites: two events with non-zero probabilities cannot be both.
- Forgetting the  $-P(A \cap B)$  and answering 1.1, which is impossible for a probability.

**Takeaway** Independent:  $P(A \cap B) = P(A)P(B)$ . Mutually exclusive:  $P(A \cap B) = 0$ . Check the numbers, not the words.

### Question 4 PRO-05 · Probability · spec M7.2

A spinner has probability 0.15 of landing on red. It is spun 200 times. What is the expected number of times it lands on red?

**Answer: A**

$$0.15 \times 200 = 30.$$

#### Full method - how and why

Expected number of successes in repeated independent trials is simply  $n \times p$ .

- 1 Identify  $n = 200$  spins and  $p = 0.15$ .  
why: Each spin is an independent trial with the same probability.
- 2 Expected number  $= 200 \times 0.15 = 30$ .  
why:  $0.15 = \frac{15}{100}$ , and 15% of 200 is 30.

#### Where it goes wrong

- Answering 15 by reading the percentage as the answer.
- Expecting the answer to be a whole number in general - expectations often are not, and that is fine.

**Takeaway** Expected frequency  $= n \times p$ .

### Question 5 PRO-06 · Probability · spec M7.7

In a class of 30, 18 study French, 12 study German and 5 study both. A student is chosen at random. Given that the student studies French, what is the probability that they also study German?

**Answer: C**

Conditional on French (18 students), 5 also study German:  $\frac{5}{18}$ .

#### Full method - how and why

The words 'given that' signal **conditional** probability: the sample space shrinks to the condition.

- 1 Restrict attention to the 18 French students - they are now the whole universe.

why: 'Given that the student studies French' means we only consider those students.

- 2 Of those 18, the number who also study German is 5.

why: The overlap is stated directly in the question.

- 3  $P(\text{German} | \text{French}) = \frac{5}{18}$   
Equivalently  $\frac{P(F \cap G)}{P(F)} = \frac{5/30}{18/30} = \frac{5}{18}$  - the 30s cancel.

#### Where it goes wrong

- Using  $\frac{5}{30}$  - the unconditional probability of studying both, ignoring the condition.
- Using  $\frac{5}{12}$ , which conditions on **German** instead - the wrong way round.

**Takeaway** Conditioning replaces the denominator: divide by the size of the group you are told you are in.

### Question 6 PRO-07 · Probability · spec M7.6

Three fair coins are tossed. Find the probability of getting more heads than tails.

**Answer: F**

Need 2 or 3 heads:  $\frac{3+1}{8} = \frac{1}{2}$  (by symmetry between heads and tails).

#### Full method - how and why

Either count the favourable outcomes out of 8, or exploit the symmetry between heads and tails.

- 1 With three coins there are  $2^3 = 8$  equally likely outcomes.

why: Each coin independently doubles the number of outcomes.

- 2 'More heads than tails' means 2 or 3 heads:  $\binom{3}{2} + \binom{3}{3} = 3 + 1 = 4$  outcomes.

why: With an odd number of coins there is no tie, so every outcome has a strict majority.

- 3  $P = \frac{4}{8} = \frac{1}{2}$ .

why: Or instantly: heads and tails are symmetric and a tie is impossible, so each must have probability  $\frac{1}{2}$ .

#### Where it goes wrong

- Answering  $\frac{3}{8}$  - the probability of exactly 2 heads, forgetting the all-heads case.
- Trying the symmetry argument with an **even** number of coins, where ties break it.

**Takeaway** An odd number of fair coins:  $P(\text{more heads}) = \frac{1}{2}$  by symmetry, since ties cannot occur.

### Question 7 PRO-09 · Probability · spec M7.4

The probabilities that a train is early, on time or late are 0.1,  $x$  and  $2x$  respectively. Find the probability that the train is not late.

**Answer: D**

$$0.1 + 3x = 1 \Rightarrow x = 0.3; \text{ not late} = 0.1 + 0.3 = 0.4.$$

#### Full method - how and why

The probabilities of an exhaustive set of outcomes sum to 1. Use that to find  $x$  first.

- 1 Early, on time, late are the only possibilities, so  $0.1 + x + 2x = 1$ .  
*why:* Exhaustive and mutually exclusive outcomes have total probability 1.
- 2 Solve:  $3x = 0.9$ , so  $x = 0.3$  and late = 0.6.  
*why:* Subtract 0.1, then divide by 3.
- 3 Not late = early + on time =  $0.1 + 0.3 = 0.4$ .  
*why:* Or use the complement:  $1 - 0.6 = 0.4$  - quicker and a useful check.

#### Where it goes wrong

- Stopping at  $x = 0.3$  and answering that, or answering 0.6, the probability of being late.
- Forgetting 'early' counts as 'not late'.

**Takeaway** Probabilities of all outcomes sum to 1; 'not late' is the complement of 'late'.

### Question 8 PRO-10 · Probability · spec M7.7

A fair die is thrown repeatedly until a six appears. Find the probability that the first six appears on the third throw.

**Answer: A**

$$\left(\frac{5}{6}\right)^2 \cdot \frac{1}{6} = \frac{25}{216}.$$

#### Full method - how and why

'First success on the  $k$ th trial' means failure, failure, ..., then success - a chain of independent events.

- 1 The first two throws must be non-sixes: probability  $\frac{5}{6}$  each.  
*why:* 'First six on the third' forbids a six on throws one and two.
- 2 The third throw must be a six: probability  $\frac{1}{6}$ .  
*why:* Exactly one way to succeed on that throw.
- 3 Multiply:  $\left(\frac{5}{6}\right)^2 \times \frac{1}{6} = \frac{25}{216}$ .  
*why:* Independent throws, so probabilities multiply.

#### Where it goes wrong

- Answering  $\frac{1}{6}$  by ignoring the first two throws.
- Using  $\left(\frac{1}{6}\right)^3 = \frac{1}{216}$ , which is the probability of **three** sixes.

**Takeaway** First success on trial  $k$ :  $(1 - p)^{k-1}p$  - failures first, then the one success.

### Question 9 PRO-11 · Probability · spec M7.6

Two cards are drawn at random from a standard pack of 52 without replacement. Which of the following expressions is the probability that both are aces?

**Answer: E**

Without replacement the second draw has 3 aces among 51 cards.

#### Full method - how and why

The question asks for the **expression**, so the reasoning is entirely about how the second draw changes.

1 First card an ace:  $\frac{4}{52}$  - four aces in the pack.  
why: All 52 cards are equally likely.

2 Given that, one ace and one card overall have gone: 3 aces remain among 51 cards.  
why: Without replacement, **both** numbers drop by one.

3 Multiply:  $\frac{4}{52} \times \frac{3}{51}$ .  
why: This is  $P(A) \times P(B | A)$ , the multiplication rule for dependent events.

#### Where it goes wrong

- $\frac{4}{52} \times \frac{4}{52}$  - that is with replacement.
- $\frac{4}{52} \times \frac{3}{52}$  - reducing the aces but forgetting the total also falls to 51.
- Adding rather than multiplying: 'and' means multiply.

**Takeaway** Without replacement, reduce numerator **and** denominator; 'and'  $\Rightarrow$  multiply.

### Question 10 PRO-12 · Probability · spec M7.7

Events  $A$  and  $B$  are mutually exclusive with  $P(A) = 0.3$  and  $P(B) = 0.4$ . Find  $P(\text{neither } A \text{ nor } B)$ .

**Answer: A**

$P(A \text{ or } B) = 0.7$ , so  $\text{neither} = 0.3$ .

#### Full method - how and why

'Neither' is the complement of 'at least one'. Mutual exclusivity makes the addition rule simple.

1 Mutually exclusive means  $P(A \cap B) = 0$ , so  $P(A \cup B) = 0.3 + 0.4 = 0.7$ .  
why: No overlap to subtract.

2 'Neither' is the complement of 'A or B':  $1 - 0.7 = 0.3$ .  
why: Everything not in  $A \cup B$ .

#### Where it goes wrong

- Answering 0.7, the probability that one of them **does** happen.
- Computing  $0.7 \times 0.6 = 0.42$  by treating them as independent - they are explicitly not, and independent events with these probabilities could not be mutually exclusive.

**Takeaway**  $\text{Neither} = 1 - P(A \cup B)$ ; and for mutually exclusive events  $P(A \cup B)$  is just the sum.

### Question 11 PRO-04 · Probability · spec M7.7

A biased coin has probability  $p$  of landing heads. It is tossed twice. The probability of exactly one head is  $\frac{3}{8}$ . Find the possible values of  $p$ .

**Answer: D**

$$2p(1-p) = \frac{3}{8} \Rightarrow 16p^2 - 16p + 3 = 0 \Rightarrow (4p-1)(4p-3) = 0.$$

#### Full method - how and why

'Exactly one head' in two tosses happens in two orders; that gives a quadratic in  $p$  with two valid roots.

1  $P(\text{exactly one head}) = P(HT) + P(TH) = p(1-p) + (1-p)p = 2p(1-p).$

why: The factor 2 is the number of orders - equivalently  $\binom{2}{1}$ .

2 Set  $2p(1-p) = \frac{3}{8}$  and clear fractions:  $16p(1-p) = 3$ , so  $16p - 16p^2 = 3$ .

why: Multiply through by 8.

3 Rearrange and factorise:  $16p^2 - 16p + 3 = 0 \Rightarrow (4p-1)(4p-3) = 0.$

why: Check:  $4 \times 4 = 16$ ,  $-1 \times -3 = 3$ , and  $-12p - 4p = -16p$  ✓.

4 So  $p = \frac{1}{4}$  or  $p = \frac{3}{4}$ ; both lie in  $[0, 1]$ , so both are valid.

why: They are mirror images - swapping heads and tails leaves 'exactly one head' unchanged, so the symmetry is expected.

#### Where it goes wrong

- Omitting the factor 2, which gives  $p^2 - p + \frac{3}{8} = 0$  with no real roots.
- Discarding one root; a biased coin can favour either face, and the symmetry  $p \leftrightarrow 1-p$  guarantees a pair.

**Takeaway** 'Exactly one' needs the count of orders; and check **both** roots against  $0 \leq p \leq 1$  rather than assuming one is spurious.

### Question 12 PRO-08 · Probability · spec M7.7

A box contains  $n$  white balls and 3 black balls. Two balls are drawn without replacement. The probability that both are white is  $\frac{5}{12}$ . Find  $n$ .

**Answer: A**

$$\frac{n}{n+3} \cdot \frac{n-1}{n+2} = \frac{5}{12} \Rightarrow 12n^2 - 12n = 5n^2 + 25n + 30 \Rightarrow 7n^2 - 37n - 30 = 0 \Rightarrow (7n+5)(n-6) = 0, \text{ so } n = 6.$$

#### Full method - how and why

Set up the without-replacement probability in terms of  $n$ , then solve the resulting quadratic.

1 There are  $n+3$  balls in total.  $P(\text{both white}) = \frac{n}{n+3} \times \frac{n-1}{n+2}.$

why: After one white is removed,  $n-1$  whites remain out of  $n+2$ .

2 Set equal to  $\frac{5}{12}$  and cross-multiply:  $12n(n-1) = 5(n+3)(n+2).$

why: Cross-multiplying is safe since all denominators are positive.

3 Expand:  $12n^2 - 12n = 5(n^2 + 5n + 6) = 5n^2 + 25n + 30.$

why: Expand the right side carefully:  $(n+3)(n+2) = n^2 + 5n + 6$ .

4 Collect:  $7n^2 - 37n - 30 = 0$ , which factorises as  $(7n+5)(n-6) = 0.$

why: Check the middle term:  $-42n + 5n = -37n$  ✓.

5  $n = -\frac{5}{7}$  is impossible for a count, so  $n = 6$ .

why: Verify:  $\frac{6}{9} \times \frac{5}{8} = \frac{30}{72} = \frac{5}{12}$  ✓.

#### Where it goes wrong

- Writing  $(\frac{n}{n+3})^2$ , the with-replacement version.
- Accepting the negative root instead of rejecting it on contextual grounds.

**Takeaway** Model without replacement by decreasing both numerator and denominator, then reject roots that cannot be counts.